

Index Coding and Fractional Graph Theory

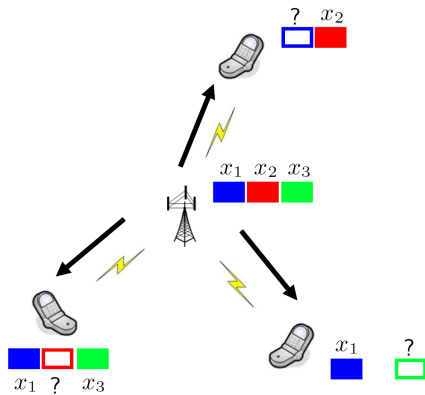
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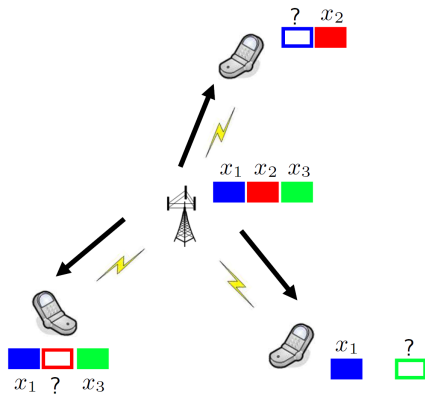
K. N. Toosi University of Technology

Index Coding Illustration



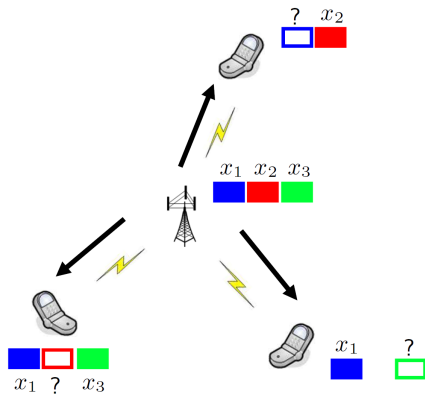
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Index Coding Illustration



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- What is the fundamental limit on the number of transmissions?

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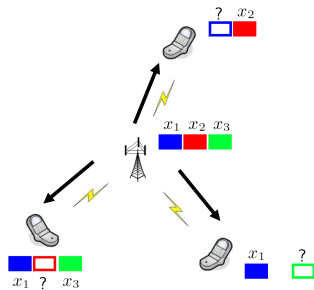


- Send $x_1 + x_2$ and x_3
- What is the fundamental limit on the number of transmissions?
- Which coding scheme achieves the limit?

Representations

- Side information

$$A_1 = \{2\}, \quad A_2 = \{1, 3\}, \quad A_3 = \{1\}$$



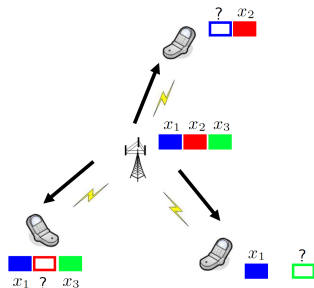
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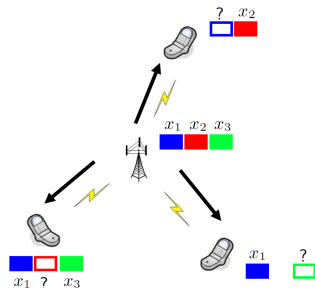
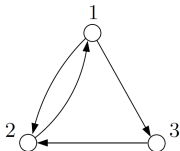
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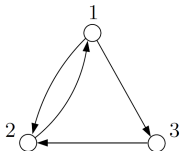
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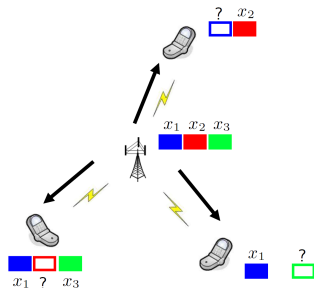
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- # of n -message index coding problems = # of n -node directed graphs



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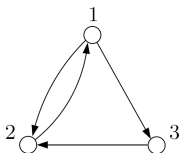
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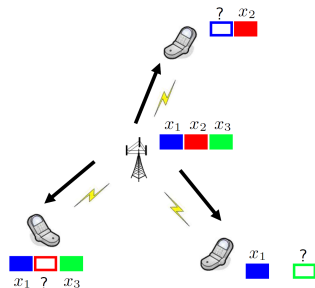
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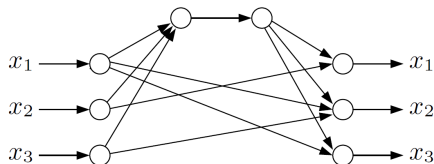
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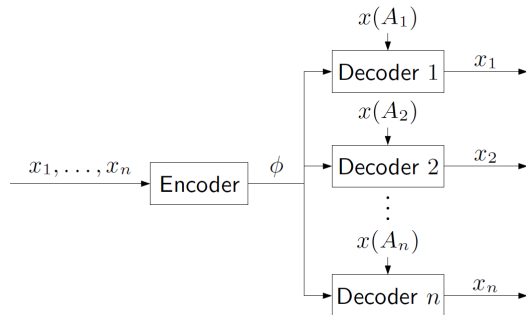
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- Multiple unicast network coding



Index coding



- $(t_1, \dots, t_n, r)$ **index code**
- **Messages** $x_j \in \{0, 1\}^{t_j}, \quad j \in [1 : n]$
- **Codeword** $\phi(x_1, \dots, x_n) \in \{0, 1\}^r$
- **Side information**

$$x(A_j) := (x_i : i \in A_j), \quad A_j \subseteq [1 : n] \setminus \{j\}$$

Achievability, Capacity region

Achievability

- A rate tuple $(R_1, \dots, R_n)$ is **achievable** iff there exists a $(t_1, \dots, t_n, r)$ **index code** such that

$$R_j \leq \frac{t_j}{r}, \quad j \in [1 : n]$$

Capacity Region

- The **capacity region** $\mathcal{C}$ is the closure of the set of all achievable rate tuples.

Broadcast Rate

- The **broadcast rate** is defined as

$$\beta = \min \left\{ b : \left(\frac{1}{b}, \dots, \frac{1}{b} \right) \in \mathcal{C} \right\}$$

- $x_j \in \{0, 1\}^t, \quad j \in [1 : n], \quad 1 \leq \beta \leq n$

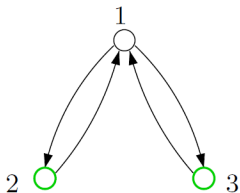
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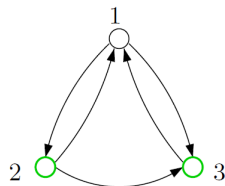
$$\beta = \min \left\{ b : \left(\frac{1}{b}, \dots, \frac{1}{b} \right) \in \mathcal{C} \right\}$$

- $x_j \in \{0, 1\}^t, \quad j \in [1 : n], \quad 1 \leq \beta \leq n$
- **No computable characterization** of the capacity region $\mathcal{C}$ or the broadcast rate β in general.
- **No approximation** within a factor of $O(n^{1-\varepsilon})$ for any $\varepsilon > 0$.
- **Existing results**
 - **Lower bounds** on the broadcast rate β
 - **Coding schemes** (upper bounds on β)

Cycle-free bound



$$2 = \alpha(G) \leq \beta$$



$$2 \leq \beta$$

Cycle-free lower bound(Neely–Tehrani–Zhang 2013)

$$\max_{S \subseteq [1:n] : G|_S \text{ is cycle-free}} |S| \leq \beta$$

Polymatroidal outer bound

Polymatroidal bound (Dougherty et al. 2011, Blasiak et al. 2011)

If $(R_1, \dots, R_n)$ is achievable for the index coding problem $(j \mid A_j), j \in [1 : n]$, then

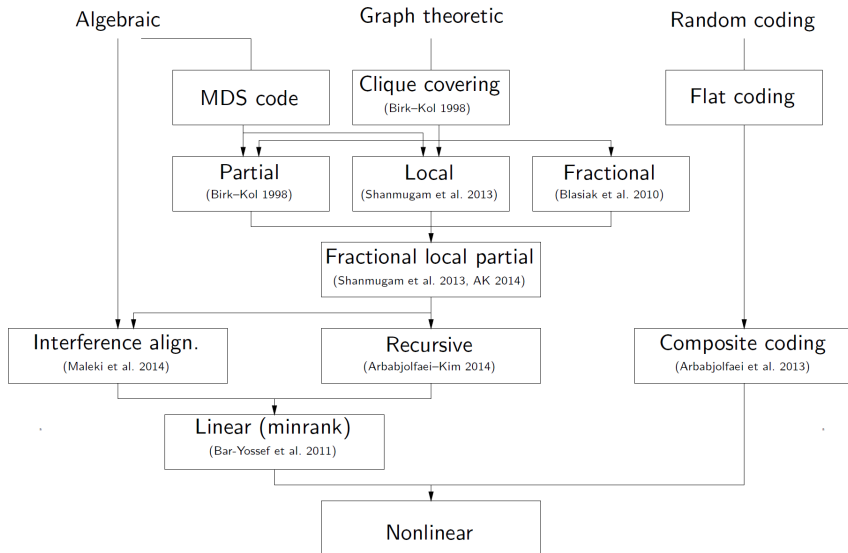
$$R_j \leq f(B_j \cup \{j\}) - f(B_j), \quad j \in [1 : n],$$

for some set function f such that

- $f(\emptyset) = 0$
- $f([1 : n]) = 1$
- $f(J) \leq f(K), \quad J \subseteq K \subseteq [1 : n]$
- $f(J \cap K) + f(J \cup K) \leq f(J) + f(K), \quad J, K \subseteq [1 : n]$

- Proof based on [submodularity of entropy](#)
- Tighter than the [cycle-free bound](#)
- Not tight (Sun–Jafar 2013)

Overview of coding schemes



MDS Code

- Side information pattern:

$$(1 \mid 2), (2 \mid 1, 3), (3 \mid 1)$$

- Every node has at least one piece of side information.
- Systematic (5, 3) MDS code:

$$(x_1, x_2, x_3, p_1, p_2)$$

- Sending only the parity symbols achieves

$$\beta = 2.$$

- In general,

$$\beta = |V| - \min_{v \in V} \text{indegree}(v).$$

- The partial clique number

$$\kappa(G) = |V| - \min_{v \in V} \text{indegree}(v).$$

- If the side information graph is a clique, then

$$\beta = 1.$$

Representing the Problem

Before we can optimize, we must model the system as a directed graph $G(V, E)$.

Graph Construction Rules:

1. **Nodes (V):** Each node represents a Receiver (who wants a specific message x_i).
2. **Edges (E):** A directed edge (i, j) exists if Receiver i **HAS** message x_j (Side Information).

$$G = (V, E)$$

Side Information Graph

Key Takeaway

The more edges we have (more side information), the fewer transmissions we need. Our goal is to exploit the structure of these edges.

Clique Covering: The Intuition

The Concept

Now, let's look at a specific structure: A **Clique**. This is when a group of receivers all know each other's desired messages.

- **Scenario:**
 - Receiver 1 wants x_1 , has x_2 .
 - Receiver 2 wants x_2 , has x_1 .
- **Solution:** Broadcast **XOR sum** ($x_1 \oplus x_2$).
- **Result:**
 - $R_1: (x_1 \oplus x_2) \oplus x_2 \rightarrow x_1$
 - $R_2: (x_1 \oplus x_2) \oplus x_1 \rightarrow x_2$
- **Cost:** 1 Transmission for 2 Users.

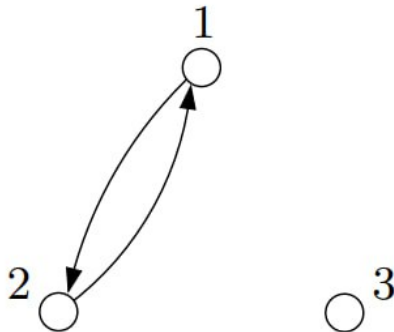


Figure: Perfect Clique

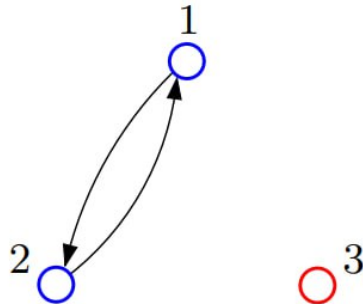
Method 1: Clique Partitioning

The Strategy

Partition the entire graph into the **minimum number of cliques**.

In this example:

1. Nodes 1 & 2 form a clique $\rightarrow$ Cost: 1.
 2. Node 3 is isolated $\rightarrow$ Cost: 1.
- **Rate (b):** Equal to the **Clique Cover Number** of the graph.
 - **Limit:** This requires bidirectional knowledge (edges going both ways).



The Limitation: The Cycle Problem

The Issue

Strict Clique Covering ignores "partial" knowledge.

Look at the cycle (1-2-3):

- 1 has 2; 2 has 3; 3 has 1.
- This is **NOT** a clique (edges are one-way).

Inefficiency:

- Standard clique algorithms treat these as separate nodes.
- **Naive Cost:** 3 Transmissions (One for each).
- We need a method that handles cycles.

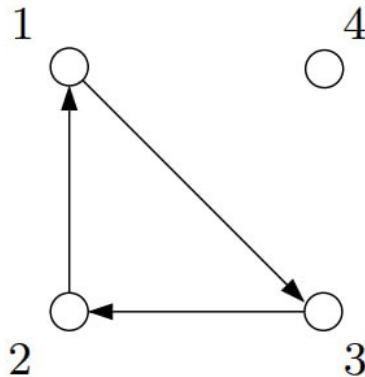


Figure: Directed Cycle (No Cliques)

Method 2: Partial Clique Covering

The Solution

Partial Clique Covering relaxes the requirement. We group nodes based on "missing" info.

- We treat the cycle (1, 2, 3) as a single group (Blue).
- Instead of simple XOR, we use **Linear Coding**.
- **Result:** We can service 3 users with fewer than 3 transmissions.
- This is equivalent to **Fractional Graph Coloring**.

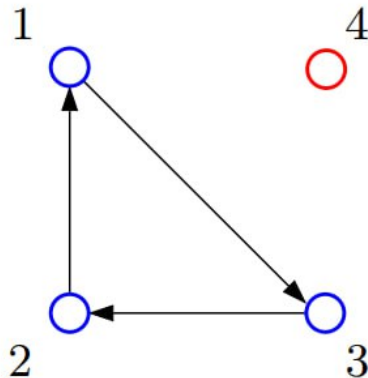


Figure: Partial Clique Grouping

Real Example: Solving the Cycle

Let's solve the Cycle (1-2-3) using just **2 transmissions** for 3 users.

1. The Setup (What users have):

- R_1 : Wants x_1 , Has x_2
- R_2 : Wants x_2 , Has x_3
- R_3 : **Wants** x_3 , **Has** x_1

2. The Transmissions (Server): We broadcast two linear sums:

$$L_1 = x_1 + x_2$$

$$L_2 = x_2 + x_3$$

3. Decoding (Perspective of R_3):

R_3 receives L_1 and L_2 . R_3 only knows x_1 .

Step-by-Step Recovery

1. Take L_1 and subtract known x_1 :

$$(x_1 + x_2) - x_1 = x_2$$

2. Now R_3 knows x_2 !

3. Take L_2 and subtract newly found x_2 :

$$(x_2 + x_3) - x_2 = x_3$$

Success! R_3 recovered x_3 .

Thank you!